Design, Implementation, and Evaluation of a AI System for Personal Memory Preservation and Retrieval

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Abstract

This work investigates retrieval-augmented generation (RAG) for personal memory preservation and retrieval through a neutral, dataset-driven evaluation. A standardized protocol is applied across MS MARCO, TimeLine QA, and OpenLifeLog QA using established retrieval (Precision/Recall, MRR, NDCG, MAP, AQWV) and generation (Exact Match, F1, ROUGE-L, semantic similarity, faithfulness) metrics with per-query significance testing. On MS MARCO, dense retrieval methods significantly outperform a BM25 baseline on ranking quality; the SBert variant attains the strongest retrieval and generation scores. On OpenLifeLog QA, a Qwen3-based configuration improves mean reciprocal rank over BM25, whereas BM25 achieves higher recall, indicating the continued utility of lexical signals in lifelog-style descriptions; the SBert variant underperforms on this dataset. These results highlight the influence of dataset characteristics on RAG behavior and the importance of aligning retrieval strategy with domain demands. The presented evaluation provides a reproducible basis for comparing RAG designs in personal information settings.

1 Introduction

1.1 Background and Motivation

RAG combines non-parametric retrieval with generative models to enhance factual accuracy and contextual relevance in question answering (Lewis et al., 2021; Karpukhin et al., 2020). While RAG has demonstrated strong performance in open-domain tasks, systematic evaluation across diverse datasets and domains remains limited.

Personal knowledge management introduces additional challenges beyond standard information retrieval. Unlike static document collections, personal memories and experiences are temporally structured, context-dependent, and often multimodal. Retrieving information from such sources requires temporal reasoning and sensitivity to individual context. Datasets such as OpenLifeLog QA (Tran et al., 2025) highlight these complexities, where queries depend on temporal and contextual cues not captured in traditional QA benchmarks.

1.2 Problem Statement and Objectives

This work aims to provide a systematic evaluation of RAG approaches for personal memory retrieval. The study focuses on how dense, embedding based strategies perform in comparison to sparse methods such as BM25 in a top-k retrieval setting. The objectives are threefold:

- **Comparative Evaluation**: Assess the performance of dense and sparse retrieval methods across datasets representing general, temporal, and personal information domains.
- **Metric-Based Analysis**: Apply standardized retrieval and generation metrics to quantify accuracy, relevance, and grounding quality in RAG responses.
- **Domain Insights**: Identify how dataset characteristics – such as temporal structure and personal context – influence retrieval and generation behavior.

2 Related Work

RAG emerged to address limitations of parametric language models in knowledge-intensive tasks. Lewis et al. (Lewis et al., 2021) introduced the foundational RAG framework, combining pre-trained parametric memory with non-parametric retrieval mechanisms, achieving state-of-the-art results on open-domain question answering (QA) while generating more factual responses than parametric-only baselines. The effectiveness of RAG systems fundamentally depends on retrieval quality, with Karpukhin et al. (Karpukhin et al., 2020) demonstrating that dense passage retrieval using learned embeddings substantially outperforms traditional sparse methods like BM25, achieving 9-19 % absolute improvements in top-20 passage retrieval accuracy.

For evaluation of these methods the MS MARCO dataset (Bajaj et al., 2018) has become a cornerstone benchmark for machine reading comprehension, comprising over one million real user queries from Bing search logs. Its scale and real-world nature distinguish it from synthetic datasets, making it valuable for practical RAG evaluation. Bonifacio et al. (Bonifacio et al., 2022) extended this with mMARCO for multilingual evaluation. Temporal reasoning presents unique challenges, with Tan et al. (Tan et al., 2023) introducing TimeLine QA to evaluate systems' ability to answer questions over temporal sequences.

Personal data QA represents a specialized domain with distinct challenges. Building on early conceptual work by Lamming (Lamming, 1994) on human memory prostheses, recent advances have produced specialized datasets. Tran et al. (Tran et al., 2022) introduced LLQA, the first standardized lifelog QA benchmark with over 15,000 questions across 85 days of data. This evolved into MemoriQA (Tran et al., 2024) and OpenLifeLog QA (Tran et al., 2025), featuring 14,187 question-answer pairs across 18 months of multimodal lifelog data. Du et al. (Du et al., 2024) contributed PerLTQA, incorporating semantic and episodic memories with emphasis on memory classification and synthesis.

Recent work has focused on optimizing RAG components and evaluation methodologies. Roychowdhury et al. (Roychowdhury et al., 2024) analyzed RAG evaluation metrics, identifying domain-specific challenges and metric interpretability issues. Yu et al. (Yu et al., 2025) conducted a systematic survey of RAG evaluation, highlighting gaps in current benchmarking approaches.

Despite significant progress, systematic comparison across diverse datasets with varying temporal and contextual requirements remains limited (Roychowdhury et al., 2024; Yu et al., 2025). The impact of specific system components (chunking strategies, embedding models, threshold filtering) on performance across different question types requires deeper investigation. This work tries to address some of these open questions by implementing a unified evaluation pipeline that systematically assesses RAG performance across MS MARCO, TimeLine QA, and OpenLifeLog QA datasets, providing insights into system design choices and their impact on different information retrieval tasks.

3 Methodology

This section presents the methodology underlying the implementation of the RAG system, the design of the evaluation pipeline used to assess its performance, and the datasets chosen for evaluation. The system is available on Github in this repository.

3.1 Retrieval Augmented Generation (RAG)

RAG combines the parametric knowledge of pre-trained language models with non-parametric external sources (Lewis et al., 2021). It retrieves relevant information from a knowledge base during generation, enhancing factual accuracy and grounding. The architecture consists of a retriever, which identifies relevant documents using dense vector representations (Karpukhin et al., 2020), and a generator, which produces responses conditioned on both the query and retrieved context.

3.1.1 RAG System Implementation

The system follows a standard RAG design with top-k retrieval and optional score thresholding. For each query, candidate passages are ranked by similarity. The top k passages are selected, and low-scoring items can be filtered by a configurable threshold.

The end-to-end flow is: (1) encode the query into a dense vector, (2) retrieve and rank documents by vector similarity (e.g., cosine), (3) concatenate the selected passages with the query to form the context, and (4) generate an answer conditioned on this context.

To prepare documents for retrieval, two complementary chunking strategies are used: fixed-length chunking by character count (a simple, size-controlled segmentation agnostic to sentence boundaries) and sentence-aware chunking (which preserves linguistic units). The former offers predictable segment sizes and the latter can improve semantic coherence at the cost of variable lengths. This combination supports analysis of trade-offs between retrieval coverage and contextual cohesion. While these strategies have been implemented and play a role in the general functionality of our software we have disabled them for evaluation since the datasets we are evaluating on are passage based instead of longer documents and therefore did not require additional chunking.

3.1.2 LLM Integration

The implementation employs Qwen3 as the primary language model, accessed through the LlamaCpp framework for efficient local deployment and the transformers library for benchmarking evaluation due to better GPU support. Rather than serving as a strict baseline, Qwen3 functions as the central generative component within the RAG system. It was selected for its strong performance on knowledge-intensive tasks, multilingual capabilities, and computational efficiency in resource-constrained environments.

The language model configuration prioritizes factual accuracy and coherent response generation, with temperature settings optimized for deterministic evaluation while maintaining response diversity. These aspects are quantitatively evaluated through generation metrics such as F1, ROUGE-L, and faithfulness scores, which measure correctness, fluency, and grounding in retrieved evidence. The system supports flexible prompt engineering, allowing for dataset-specific instruction formatting and context integration strategies. This ensures reproducible results while maintaining the model's ability to generate contextually appropriate responses across diverse question types.

3.1.3 Vector Storage and Retrieval

The vector storage solution utilizes Qdrant, a high-performance database for similarity search, chosen for its local deployment capabilities, efficient in-memory operations, and flexible integration with a wide range of embedding models through a model-agnostic API design.

Depending on the use case, the system operates in persistent or in-memory mode, enabling flexible deployment from development to production. Before embedding generation, documents can be segmented through configurable chunking strategies, supporting comparative evaluation of different representation methods.

Similarity search relies on cosine similarity optimized for dense vectors, enhanced through approximate nearest neighbor indexing. Metadata stored with each vector enables filtered search and provenance tracking, ensuring efficient and accurate retrieval across large collections while supporting detailed performance analysis.

3.2 Evaluation Pipeline

The evaluation pipeline implements a comprehensive automated framework designed to systematically assess RAG system performance across multiple datasets and configurations. The framework architecture centers on a RAG evaluation component, which orchestrates the complete evaluation workflow from dataset loading through metric computation and result persistence.

The pipeline supports multiple evaluation configurations through modular design, enabling comparison between baseline approaches and optimized RAG implementations. The framework implements two primary evaluation modes: baseline configurations using traditional sparse retrieval methods (BM25) and advanced configurations employing dense vector representations with neural embedding models. This dual-mode approach facilitates direct comparison between established information retrieval techniques and modern RAG architectures.

3.2.1 Metric Selection and Implementation

The evaluation framework implements a comprehensive suite of metrics addressing both retrieval effectiveness and generation quality(Yu et al., 2025). Retrieval metrics focus on the system's ability to identify relevant documents, employing standard information retrieval measures including precision, recall, F1-score, Mean Reciprocal Rank (MRR), and Normalized Discounted Cumulative Gain (NDCG). These metrics are implemented in a retrieval metrics module, providing standardized calculation procedures across all datasets.

The quality of generated responses is assessed using multiple complementary metrics. Traditional n-gram overlap measures, such as ROUGE-L, capture surface-level similarity between generated and reference answers. Additional metrics include exact match accuracy for factual questions and answer completeness measures for complex queries requiring multi-faceted responses. For further assesment semantic similarity measures (SBert similarity, cross-encoder scoring) are also incorporated to partially capture subjective relevance.

3.2.2 Baseline Comparison Methodology

The evaluation framework establishes rigorous baseline comparisons through traditional sparse retrieval methods implemented within dedicated baseline components. The BM25 baseline, realized through a Lucene-based indexing system, provides a strong and widely recognized reference point for lexical matching approaches. This baseline operates with retrieval-only generation (top-1 retrieval), returning the highest-scoring passage according to BM25 as the final answer.

Baseline configurations use the same preprocessing pipeline and evaluation metrics to ensure a fair comparison. The sparse retrieval component preserves identical document processing steps but replaces dense embeddings with classical term-frequency–inverse document frequency (TF–IDF) representations. This consistency allows observed performance differences to be attributed to architectural factors rather than implementation discrepancies.

3.2.3 Statistical Significance Testing and Performance

The evaluation framework supports statistical analysis by exporting detailed, per-query results that can be compared across configurations for significance testing. Response times are recorded during evaluation to approximate computational efficiency, complementing accuracy metrics for a more complete performance profile.

During each evaluation run, the framework logs retrieval counts, processing times, and intermediate metric values. These data are stored in structured JSON reports that include per-query results, aggregate statistics, and full system configurations. This structure enables detailed analysis and exact reproduction of evaluation outcomes.

3.3 Datasets

This subsection presents the datasets used to evaluate the RAG system, selected to cover both general-domain and specialized question-answering tasks. The chosen datasets: MS MARCO, TimeLine QA, and OpenLifeLog QA, enable a comprehensive assessment of retrieval and generation performance across diverse contexts, ranging from web-based information retrieval to temporally grounded and personal memory scenarios.

3.4 MS Marco

The MS MARCO (Microsoft MAchine Reading COmprehension) dataset (Bajaj et al., 2018) serves as a foundational benchmark for machine reading comprehension and information retrieval tasks. The dataset comprises over one million queries derived from real Bing search logs, paired with human-generated answers and extracted web passages. This real-world provenance distinguishes MS MARCO from synthetic datasets, providing authentic user information needs and query formulations.

The dataset structure consists of three primary components: queries representing actual user search intents, a comprehensive document collection extracted from web sources, and relevance judgments (qrels) that establish ground truth associations between queries and relevant passages. Query complexity ranges from simple factoid questions to moderately compositional queries that may require integrating information across multiple passages.

There are multiple versions specializations for this dataset. For our evaluation the passage v1 dataset with the dev question set was used to ensure we can provide results that are as comparable as possible. This version does not include human annotated gold answers and therefore has been mainly used for evaluation of the retrieval performance. Additionally the MS MARCO QnA dataset was used with a sampled 10’000 question subset, because there is no dev split provided. This dataset was then used for both retrieval and generation evaluation.

3.5 TimeLine QA

Timeline QA addresses the specialized domain of temporal question answering, where queries require understanding of chronological relationships and time-dependent information (Tan et al., 2023). The dataset presents unique challenges distinct from traditional QA benchmarks, emphasizing temporal reasoning capabilities essential for personal memory and lifelog applications. This dataset is generated using python scripts and allows for three different densities (sparse, medium, dense) which represent different amounts of events the needle (haystack) is hidden in, with dense being the most difficult version.

The dataset encompasses diverse temporal question types, including sequence ordering, duration calculation, temporal containment, and causal relationships across time periods. Questions span multiple temporal granularities, from precise timestamps to broader temporal intervals, requiring systems to handle both explicit and implicit temporal references. The temporal coverage spans realistic lifelog scenarios, incorporating daily activities, recurring events, and long-term trends that characterize personal information management tasks.

Evaluation criteria for Timeline QA extend beyond traditional accuracy metrics to include temporal coherence and chronological consistency. For integration with the evaluation pipeline, each event description is augmented with its associated timestamp, allowing the system to retrieve passages that are temporally aligned with the query. This simple temporal annotation supports basic time-aware retrieval without explicit reasoning over temporal relations.

3.6 OpenLifeLog QA

OpenLifeLog QA focuses on personal information retrieval within lifelogging contexts (Tran et al., 2025). It captures real-world personal activities and experiences through multimodal data – including images, locations, and textual event descriptions – paired with annotated questions and answers.

The personal information retrieval context presents distinct challenges compared to general-domain QA tasks. Questions often require understanding of personal context that is not readily available in public knowledge bases. The dataset's lifelogging characteristics include temporal dependencies, location-based queries, activity recognition, and social interaction patterns that demand sophisticated context modeling and retrieval strategies.

Privacy considerations play a crucial role in the datasets' design and handling procedures. All personal identifiers undergo anonymization, and data access follows strict privacy protocols to ensure participant confidentiality (Tran et al., 2025). The dataset structure maintains essential contextual information while protecting individual privacy through systematic de-identification processes.

In this evaluation, only the textual components of OpenLifeLog QA are used, with event descriptions treated as passages for retrieval and question–answer pairs serving as evaluation instances. Evaluation metrics adaptation accounts for the subjective nature of personal information retrieval, where multiple valid answers may exist for individual queries. The integration framework employs established QA evaluation metrics including exact match and F1 scoring for objective assessment, while incorporating semantic similarity measures (SBert similarity, cross-encoder scoring) to partially capture subjective relevance. However, the framework currently lacks specialized metrics for personal context appropriateness and temporal consistency that would be ideal for comprehensive personal memory retrieval evaluation.

4 Results

4.1 MS MARCO Dataset

The MS MARCO evaluation comprises two complementary assessments: the passage-ranking subset and the QnA subset. Each query includes annotated relevance judgments linking it to web passages, enabling controlled comparisons across retrieval and generation configurations.

4.1.1 Retrieval Performance Comparison

The evaluation on the MS MARCO dataset reveals significant differences in retrieval performance across the three configurations. Table 1 presents retrieval metrics for the passage-ranking evaluation, while Table 2 provides comprehensive retrieval and generation metrics for the QnA evaluation, demonstrating the effectiveness of dense embedding approaches over traditional sparse retrieval methods.

Table 1: MS MARCO Passage Retrieval Metrics Comparison

Metrics	BM25	SBert	Qwen3
Precision@20	0.0244	**0.0344**	0.0262
Recall@1	0.0968	**0.1840**	0.1177
Recall@5	0.2780	**0.4437**	0.3084
Recall@10	0.3738	**0.5613**	0.4062
Recall@20	0.4681	**0.6566**	0.5010
F1 Score	0.0463	**0.0652**	0.0497
MRR	0.1849	**0.3088**	0.2111
NDCG@10	0.2229	**0.3614**	0.2502
MAP	0.1814	**0.3037**	0.2069

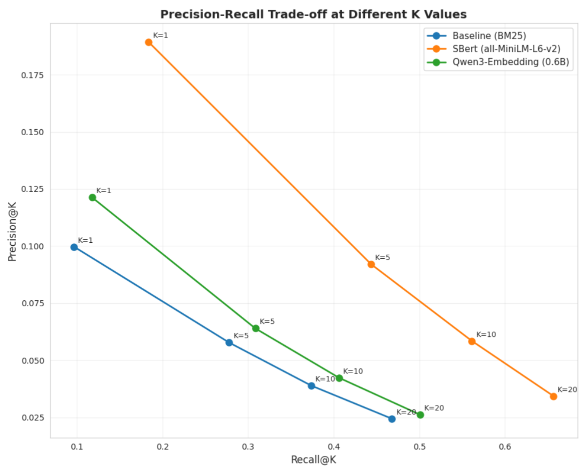

Figure 1: Precision@K and Recall@K performance across retrieval depths. SBert consistently outperforming both other contenders.

On the passage-ranking evaluation, SBert achieves the strongest performance across all retrieval metrics, with an MRR of 0.3088 representing a 67% improvement over the BM25 baseline (0.1849). The Qwen3 configuration also outperforms BM25, achieving an MRR of 0.2111 (14% improvement). Figure 1 illustrates the precision-recall trade-off analysis, revealing that dense embedding methods maintain higher precision while significantly improving recall, with SBert achieving 0.6566 recall@20 compared to 0.4681 for BM25 (40% improvement).

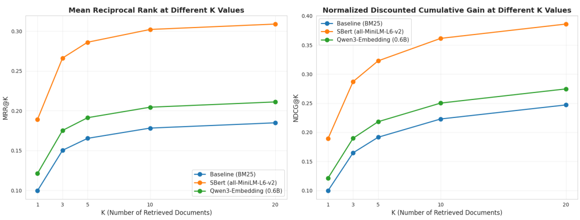

Figure 2: MRR@K and NDCG@K performance showing consistent superiority of dense embedding approaches.

The ranking quality improvements are further demonstrated in Figure 2, which shows MRR and NDCG performance across different retrieval depths (K=1,5,10,20). Both metrics consistently favor the dense embedding configurations, with the most pronounced improvements observed at lower K values, indicating better ranking of highly relevant documents.

4.1.2 Statistical Significance Analysis

Paired Approximate Randomization (PAR) tests and Wilcoxon signed-rank tests were applied to evaluate statistical significance of performance differences. SBert versus BM25 shows statistically significant differences ($p < 0.001$) across all metrics, with 67% improvement in MRR and 62% in NDCG@10. Qwen3 versus BM25 demonstrates statistically significant differences ($p < 0.001$) with 14% improvement in MRR and 12% in NDCG@10. Qwen3 versus SBert comparison yields statistically significant differences ($p < 0.001$), with SBert achieving 32% higher MRR and 31% higher NDCG@10.

4.1.3 Generation Quality Assessment

MS MARCO generation metrics reveal complementary patterns to retrieval performance. SBert achieves the strongest generation performance with an F1 score of 0.3660 and ROUGE-L score of 0.3253, substantially outperforming both BM25 (F1: 0.1994, ROUGE-L: 0.1726) and Qwen3 (F1: 0.3092, ROUGE-L: 0.2751). These generation improvements directly reflect the superior retrieval performance observed for dense embedding methods, as better retrieval provides more relevant context for answer generation. Figure 3 provides a comprehensive visualization of generation performance across all evaluated dimensions, demonstrating how improved retrieval quality translates to enhanced answer quality, support metrics, and semantic similarity measures.

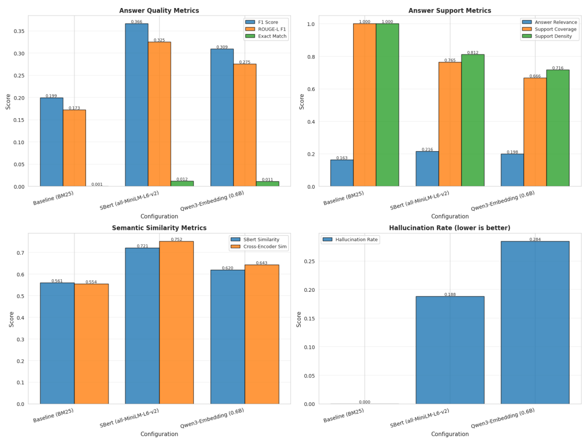

Figure 3: Generation metrics comparison

Semantic similarity metrics further confirm SBert's advantage, achieving SBert similarity of 0.7207 and cross-encoder similarity of 0.7518, compared to BM25's 0.5605 and 0.5543 respectively. Support coverage is highest for SBert at 0.7647, while Qwen3 achieves 0.6659, indicating that stronger retrieval enables the generation model to better ground responses in retrieved passages. Hallucination rates are relatively low across dense configurations (0.1879 for SBert, 0.2844 for Qwen3), though the BM25 baseline shows minimal hallucination due to its retrieval-only generation approach (see Table 2).

4.2 OpenLifeLog QA Dataset

4.2.1 Personal Information Retrieval Performance

OpenLifeLog QA evaluation shows different performance patterns across configurations (see Table 3). Qwen3 achieves MRR of 0.2850, representing a 3% improvement over BM25 baseline (0.2768), though this difference is not statistically significant ($p = 0.4468$). BM25 demonstrates the strongest recall performance with recall@20 of 0.5071, compared to Qwen3's 0.4105 and SBert's 0.1463. Figure 4 illustrates the precision-recall trade-off across different retrieval depths.

Table 2: MS MARCO-QnA Retrieval and Generation Metrics Comparison

Metrics	**BM25**	**SBert**	**Qwen3**
Retrieval Metrics			
Recall@1	0.0407	**0.0413**	0.0328
Recall@5	0.1441	**0.1449**	0.1113
Recall@10	0.2070	**0.2107**	0.1622
Recall@20	0.2676	**0.2783**	0.2186
Precision@20	0.1338	**0.1391**	0.1093
F1 Score (@20)	0.1784	**0.1855**	0.1457
MRR	0.5043	**0.5226**	0.4253
NDCG@10	0.1645	**0.2633**	0.1760
MAP	0.1699	**0.1679**	0.1253
Generation Metrics			
F1 Score	0.1994*	**0.3660**	0.3092
ROUGE-L F1	0.1726*	**0.3253**	0.2751
Support Coverage	1.0000*	0.7647	0.6659
Hallucination Rate	0.0000*	0.1879	0.2844
SBert Similarity	0.5605*	**0.7207**	0.6196
Cross-Encoder Sim	0.5543*	**0.7518**	0.6431
Efficiency Metrics			
Avg. Response Time (s)	0.0172*	1.5599	1.4492

* starred values denote top-1 retrieval generation baseline not directly comparable to generative models.

Table 3: OpenLifeLog QA Retrieval and Generation Metrics Comparison

Metrics	**BM25**	**SBert**	**Qwen3**
Retrieval Metrics			
Recall@1	0.1280	0.0346	**0.1379**
Recall@5	**0.2998**	0.0777	0.2748
Recall@10	**0.3983**	0.1072	0.3397
Recall@20	**0.5071**	0.1463	0.4105
Precision@20	**0.0440**	0.0119	0.0359
F1 Score (@20)	**0.0770**	0.0214	0.0634
MRR	0.2768	0.0770	**0.2850**
NDCG@10	**0.2814**	0.0730	0.2619
MAP	**0.2369**	0.0592	0.2201
AQWV	**0.4793**	0.1176	0.3825
Generation Metrics			
F1 Score	0.1276*	0.1487	**0.1759**
ROUGE-L F1	0.1146*	0.1472	**0.1725**
Answer Relevance	0.3420*	0.1557	0.2494
Support Coverage	1.0000*	0.3062	0.4331
Support Density	1.0000*	0.2512	0.4044
Hallucination Rate	0.0000*	0.7488	0.5956
SBert Similarity	0.2770*	0.2065	0.2751
Cross-Encoder Sim	0.0495*	0.0313	**0.0876**
Efficiency Metrics			
Avg. Response Time (s)	0.0054*	0.4556	0.7555

* starred values denote top-1 retrieval generation baseline not directly comparable to generative models.

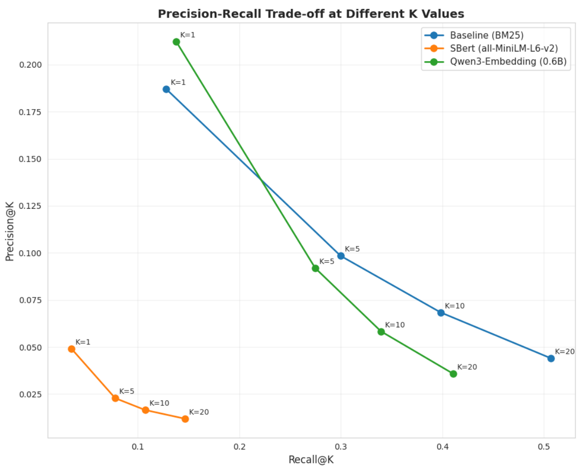

Figure 4: Precision@K and Recall@K performance for OpenLifeLog QA across retrieval depths. BM25 achieves highest recall while Qwen3 maintains competitive precision.

SBert shows substantially lower performance with MRR of 0.0770 (72% lower than baseline, $p < 0.001$) and NDCG@10 of 0.0730 (74% lower than baseline, $p < 0.001$). Qwen3 maintains NDCG@10 of 0.2619 compared to baseline's 0.2814 ($p = 0.0439$). Figure 5 shows MRR and NDCG performance across different K values, demonstrating BM25 and Qwen3's comparable ranking quality.

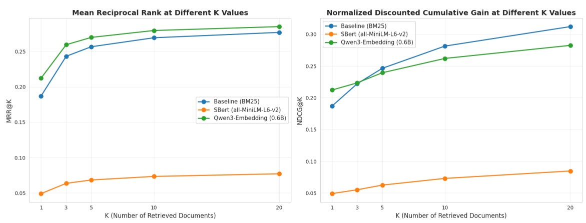

Figure 5: MRR@K and NDCG@K performance for OpenLifeLog QA showing BM25 and Qwen3 maintaining comparable ranking quality while SBert underperforms.

4.2.2 Statistical Significance Analysis

Paired Approximate Randomization (PAR) tests show varied significance levels across configurations. Qwen3 versus BM25: precision, recall, NDCG@20, and F1 show statistical significance

($p < 0.05$), while MRR shows no significant difference ($p = 0.4468$). NDCG@10 shows marginal significance ($p = 0.0439$).

SBert versus BM25 demonstrates statistically significant differences ($p < 0.001$) across all retrieval metrics, with SBert showing 72% lower MRR and 74% lower NDCG@10. Qwen3 versus SBert yields statistically significant differences ($p < 0.001$) across all metrics, with Qwen3 achieving 270% higher MRR and 259% higher NDCG@10.

4.2.3 Generation Quality Assessment

Qwen3 achieves the strongest generation performance with F1 score of 0.1759 and ROUGE-L score of 0.1725, followed by SBert (F1: 0.1487, ROUGE-L: 0.1472) and BM25 baseline (F1: 0.1276, ROUGE-L: 0.1146). Support coverage is 0.4331 for Qwen3 and 0.3062 for SBert, with support density of 0.4044 and 0.2512 respectively. Figure 6 provides a comprehensive visualization of generation performance across all dimensions.

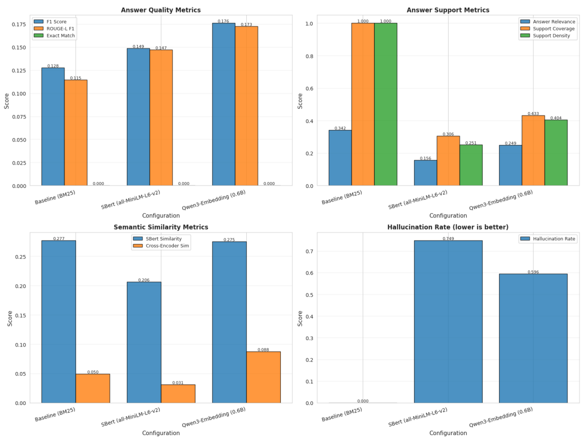

Figure 6: Generation metrics comparison for OpenLifeLog QA showing Qwen3's superiority across answer quality, support, and similarity dimensions, with notably higher hallucination rates for dense configurations.

Hallucination rates are 0.5956 for Qwen3 and 0.7488 for SBert, substantially higher than observed in MS MARCO evaluation. BM25 maintains zero hallucination rate through its retrieval-only generation approach. Semantic similarity metrics show Qwen3 with cross-encoder similarity of 0.0876 and SBert similarity of 0.2751, compared to BM25's 0.0495 and 0.2770 respectively. Answer relevance is highest for BM25 at 0.3420, followed by Qwen3 at 0.2494 and SBert at 0.1557 (see Table 3).

4.3 Timeline QA Dataset Results

Timeline QA evaluation uses the medium configuration subset, presenting temporal reasoning challenges distinct from general-domain retrieval. This dataset has very specific question-answer structures and wording due to being a purely generated dataset. Thus BM25 achieves the strongest retrieval performance with MRR of 0.5901, followed by Qwen3 (0.4767, 19% lower, $p < 0.001$) and SBert (0.1929, 67% lower, $p < 0.001$). BM25 demonstrates highest recall@20 of 0.9006 compared to Qwen3's 0.7446 and SBert's 0.3796. NDCG@10 follows similar patterns with BM25 at 0.6232, Qwen3 at 0.5169 (17% lower, $p < 0.001$), and SBert at 0.2181 (65% lower, $p < 0.001$). Statistical significance testing confirms all differences are statistically significant across all retrieval metrics (see Table 4).

Table 4: Timeline QA (Medium Configuration) Retrieval Metrics Comparison

Metrics	BM25	SBert	Qwen3
Recall@1	**0.4882**	0.1390	0.3864
Recall@5	**0.6818**	0.2542	0.5810
Recall@10	**0.7604**	0.3139	0.6634
Recall@20	**0.9006**	0.3796	0.7446
Precision@20	**0.0450**	0.0190	0.0372
F1 Score (@20)	**0.0858**	0.0362	0.0709
MRR	**0.5901**	0.1929	0.4767
NDCG@10	**0.6232**	0.2181	0.5169
MAP	**0.5901**	0.1929	0.4767
AQWV	**0.8770**	0.3553	0.7208

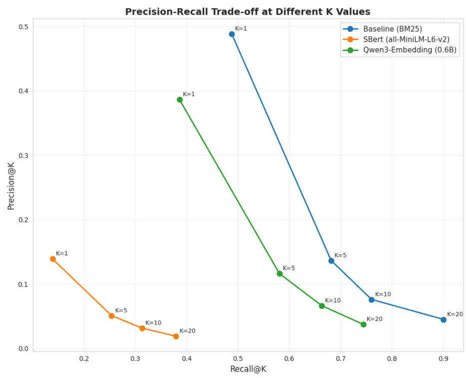

Figure 7: Precision-Recall trade-off for Timeline QA (medium configuration) showing BM25's superior performance across all retrieval depths for temporal queries.

The precision-recall trade-off reveals that BM25 maintains superior performance across all K values for temporal queries, with dense embeddings showing reduced effectiveness. Figure 7 illustrates this pattern, demonstrating BM25's strong precision at K=1 (0.4882) compared to Qwen3 (0.3864) and SBert (0.1390), with the performance gap widening at higher K values.

5 Analysis

5.1 MS MARCO Evaluation

Retrieval and generation performance are evaluated on two MS MARCO subsets: the passage ranking subset (passage) and the question answering subset (QnA). A lexical BM25 baseline with two dense retrievers are compared: SBert (`all-MiniLM-L6-v2`) and Qwen3-Embedding (0.6B).

Dense retrieval outperforms BM25 on both passage and QnA evaluations. SBert achieves the highest retrieval effectiveness overall, showing improvements of approximately +67% MRR and +56–62% NDCG over BM25 on passage ranking, while Qwen3-Embedding also yields clear gains (+14% MRR and +11–12% NDCG). This result is notable because Qwen3-Embedding is a more recent and generally stronger model, yet SBert performs better in this setting. All improvements over BM25 are statistically significant ($p < 0.05$).

On the QnA evaluation, SBert again ranks highest (MRR 0.5226 vs. 0.5043 for BM25), with Qwen3-Embedding at 0.4253. Dataset size is not the determining factor here; the key difference is the retrieval objective. The relative ordering (SBert $>$ Qwen3 $>$ BM25) remains stable across recall@K values.

Generation performance follows always retrieval performance. Dense retrieval provides more relevant supporting passages, leading to higher F1 and ROUGE-L scores. SBert yields the best generation quality (F1 $= 0.366$, ROUGE-L $= 0.325$). Qwen3-Embedding produces more complete answers than BM25 but shows slightly higher hallucination rates. BM25 exhibits low hallucination because it returns a retrieved passage verbatim, but it lacks answer completeness and grounding.

Measured latencies are not directly comparable because BM25 does not perform generation; it only returns the top retrieved passage. The majority of latency in the dense setups comes from the LLM answer generation step, not from retrieval itself. Thus, while dense retrieval incurs higher end-to-end runtime, this reflects the added generation component rather than inefficiency in the embedding models.

5.2 OpenLifeLog QA Evaluation

The OpenLifeLog QA results differ notably from the MS MARCO findings. In this setting, Qwen3-Embedding performs substantially better than SBert, which is consistent with expectations for models optimized for richer semantic matching. Qwen3-Embedding achieves a higher MRR than BM25 (0.2850 vs. 0.2768), capturing contextual cues in personal and event-centric descriptions. However, BM25 attains higher recall (0.5071 vs. 0.4105), indicating that lexical cues (e.g., activity names, dates, locations) remain a strong signal in lifelog retrieval. In practice, this makes BM25 competitive, simpler to deploy, and faster to query, raising a meaningful question about whether dense embeddings alone are sufficient for personal memory retrieval. The results suggest that a hybrid lexical+dense retriever may be advantageous in this domain.

SBert underperforms on this dataset, showing unstable ranking behavior and reduced recall across depths. Since SBert is trained on generic sentence similarity, it does not effectively capture personal, temporal, and routine-specific cues present in lifelog data. Consequently, it retrieves weaker supporting evidence, which limits its downstream performance.

Generation quality aligns with retrieval quality: Qwen3-Embedding provides more contextually appropriate retrieved evidence and therefore yields more coherent and relevant RAG outputs. BM25-based answers are more literally grounded in retrieved text but often lack completeness. SBert produces the weakest generations because its retrieval stage fails to supply adequate memory evidence.

The reported response times (1.8–2.0 s) refer to the time required to generate the complete answer. In practice, the first token appears earlier, so interactive latency is lower. Overall, these findings indicate that while dense retrieval helps, lexical signals remain highly valuable in personal lifelog settings, and hybrid retrieval approaches are a promising direction.

5.3 TimeLine QA Evaluation

The TimeLine QA results are less informative than the other evaluations because the dataset is synthetic and each entry includes an explicit date. Under these conditions, BM25 has a strong advantage: simply matching the date token often retrieves the correct event, making lexical retrieval highly effective and easy to apply. This also means that dense embeddings do not provide clear improvements here. Qwen3-Embedding performs slightly better than SBert, but neither meaningfully outperforms BM25 because semantic similarity is not the primary retrieval signal in this task.

Generation quality mirrors retrieval. When BM25 retrieves the correct dated entry, the final answer is accurate; when the retrieval is incorrect, generation fails regardless of the model. The reported times refer to complete answer generation, not time-to-first-token.

Overall, TimeLine QA in its current form favors lexical retrieval due to the explicit date structure. A more informative extension would be to regenerate the dataset with greater variability in phrasing and event descriptions, reducing near-duplicate question–answer pairs and allowing semantic retrieval methods to play a more meaningful role.

5.4 Limitations

Several limitations constrain the generalizability of these findings. First, the evaluation relies on specific datasets – MS MARCO, TimeLine QA, and OpenLifeLog QA – that do not fully align with personal memory retrieval in scope or modality. This limits direct transferability to broader domains or multimodal lifelog applications. Although we aimed to build a robust validation pipeline by separating generative performance from retrieval, generation still depended on the quantity and quality of retrieved passages. Thus, weaker retrieval directly reduced generative performance. Ideally, retrieval and generation should be evaluated both independently and jointly. Second, the absence of human evaluation means the reported metrics primarily capture surface-level correctness rather than nuanced interpretability or subjective relevance. Third, the study uses only one language model family (Qwen3) without exploring cross-model variability, which may bias generation performance. Finally, retrieval parameters such as top-k and threshold values were fixed across datasets for comparability, potentially overlooking dataset-specific optimizations.

6 Conclusion

This work presented the design and implementation of a retrieval-augmented generation system for personal memory preservation and retrieval, together with a unified, dataset-driven evaluation pipeline. We compared a sparse lexical baseline (BM25) with two dense embedding-based configurations (SBert and Qwen3-Embedding) across MS MARCO, OpenLifeLog QA and TimeLine QA. The same protocol and metric set were applied throughout, enabling controlled, per-query comparisons of retrieval quality and associated generation behaviour.

Our results show that there is no single universally superior retrieval strategy. On MS MARCO, dense retrieval substantially outperforms BM25 on standard IR metrics, and this stronger evidence generally coincides with improved answer quality. In contrast, on OpenLifeLog QA, a personal lifelog-style dataset, BM25 achieves higher recall and stronger overall ranking metrics, while Qwen3-Embedding modestly improves MRR and certain generation scores. This suggests that lexical cues such as activity names, locations, and timestamps remain highly informative in personal memory settings. TimeLine QA, being synthetic and heavily date-driven, further highlights how strongly structured temporal signals can favour simple lexical matching, but we regard these findings as less representative for real-world memory retrieval.

Taken together, these observations argue against treating dense retrieval as a drop-in replacement for lexical methods in personal memory applications. Instead, they point toward hybrid designs that deliberately combine lexical and dense signals, e.g. via parallel retrieval and fusion, lexical pre-filtering followed by dense reranking, or model-switching strategies conditioned on query type. Such combinations are particularly promising for memory retrieval, where the fine-grained, user-specific language of lifelogs and notes often aligns well with lexical scoring, while dense embeddings can capture paraphrases and broader semantic relations.

Future work should therefore focus on systematically exploring hybrid retrieval architectures in lifelog and personal knowledge domains, extending the evaluation to multimodal signals and richer temporal models, and incorporating human-centric measures of usefulness and trust. Our evaluation pipeline provides a reproducible starting point for these investigations and underscores the importance of aligning retrieval strategy with the specific characteristics of personal memory data rather than relying on results from general web-scale benchmarks alone.

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Appendix

A Detailed Evaluation Results

This appendix provides comprehensive evaluation results from all datasets, including detailed statistical tables and visualizations that complement the summary presented in the Results section.

This appendix is added for the sake of completeness and traceability of our experiments. Unlike the subset of graphs and tables included in the content of the report, they are intended to be interpreted by the readers themselves.

Each subsection of the data set follows the same structure to improve clarity. We present identical tables and plots for all datasets, generated by our evaluation pipeline and Jupyter notebook.

A.1 MS MARCO Passage

The MS MARCO passage evaluation assesses retrieval performance on 6,980 queries from the development set. This section provides comprehensive results including global metrics, statistical significance testing, and visualizations.

A.1.1 Global Retrieval Metrics

Table 5 presents the complete set of retrieval metrics across all three configurations.

Table 5: MS MARCO Passage: Global Retrieval Metrics Comparison

Metric	**BM25**	**SBert**	**Qwen3**
Precision@20	0.0244	0.0344	0.0262
Recall@1	0.0968	0.1840	0.1177
Recall@5	0.2780	0.4437	0.3084
Recall@10	0.3738	0.5613	0.4062
Recall@20	0.4681	0.6566	0.5010
F1 Score	0.0463	0.0652	0.0497
MRR	0.1849	0.3088	0.2111
NDCG@10	0.2229	0.3614	0.2502
NDCG@20	0.2554	0.3966	0.2848
MAP	0.1814	0.3037	0.2069
AQWV	0.4302	0.6125	0.4713

Table 6 shows the percentage improvements over the BM25 baseline across all metrics.

Table 6: MS MARCO Passage: Percentage Improvements Over BM25 Baseline

Metric	**SBert (%)**	**Qwen3 (%)**
Precision@20	+40.98	+7.38
Recall@1	+90.08	+21.59
Recall@5	+59.60	+10.94
Recall@10	+50.16	+8.67
Recall@20	+40.25	+7.03
F1 Score	+40.82	+7.34
MRR	+67.01	+14.17
NDCG@10	+62.13	+12.25
NDCG@20	+55.29	+11.51
MAP	+67.41	+14.06
AQWV	+42.38	+9.56

A.1.2 Statistical Significance Testing

Table 7 presents detailed statistical significance testing results for SBert vs BM25 using Paired Approximate Randomization (PAR) and Wilcoxon signed-rank tests.

Table 7: MS MARCO Passage: SBert vs BM25 Statistical Significance

Metric	SBert	BM25	Diff	Diff (%)	PAR p	Sig.
Precision	0.0652	0.0463	+0.0189	+40.82	<0.001	Yes
Recall	0.6566	0.4681	+0.1885	+40.27	<0.001	Yes
MRR	0.3088	0.1849	+0.1239	+67.01	<0.001	Yes
NDCG@10	0.3614	0.2229	+0.1385	+62.13	<0.001	Yes
NDCG@20	0.3966	0.2554	+0.1412	+55.29	<0.001	Yes
F1	0.0652	0.0463	+0.0189	+40.82	<0.001	Yes

Table 8 presents statistical significance testing results for Qwen3 vs BM25.

Table 8: MS MARCO Passage: Qwen3 vs BM25 Statistical Significance

Metric	Qwen3	BM25	Diff	Diff (%)	PAR p	Sig.
Precision	0.0497	0.0463	+0.0034	+7.34	<0.001	Yes
Recall	0.5010	0.4681	+0.0329	+7.03	<0.001	Yes
MRR	0.2111	0.1849	+0.0262	+14.17	<0.001	Yes
NDCG@10	0.2502	0.2229	+0.0273	+12.25	<0.001	Yes
NDCG@20	0.2848	0.2554	+0.0294	+11.51	<0.001	Yes
F1	0.0497	0.0463	+0.0034	+7.34	<0.001	Yes

Table 9 presents statistical significance testing results for Qwen3 vs SBert.

Table 9: MS MARCO Passage: Qwen3 vs SBert Statistical Significance

Metric	Qwen3	SBert	Diff	Diff (%)	PAR p	Sig.
Precision	0.0497	0.0652	-0.0155	-23.77	<0.001	Yes
Recall	0.5010	0.6566	-0.1556	-23.70	<0.001	Yes
MRR	0.2111	0.3088	-0.0977	-31.64	<0.001	Yes
NDCG@10	0.2502	0.3614	-0.1112	-30.77	<0.001	Yes
NDCG@20	0.2848	0.3966	-0.1118	-28.19	<0.001	Yes
F1	0.0497	0.0652	-0.0155	-23.77	<0.001	Yes

A.1.3 Performance Visualization

Following are the evaluation graphs allowing for easier overview and evaluation of the many of the values already present in the preceding tables. For completeness we include all generated plots of the obtained results in the appendix for interested readers to interpret them themselves. The plots deemed most relevant are analyzed in the reports result section above, outside of the appendix.

Figure 8 shows precision and recall at different K values.

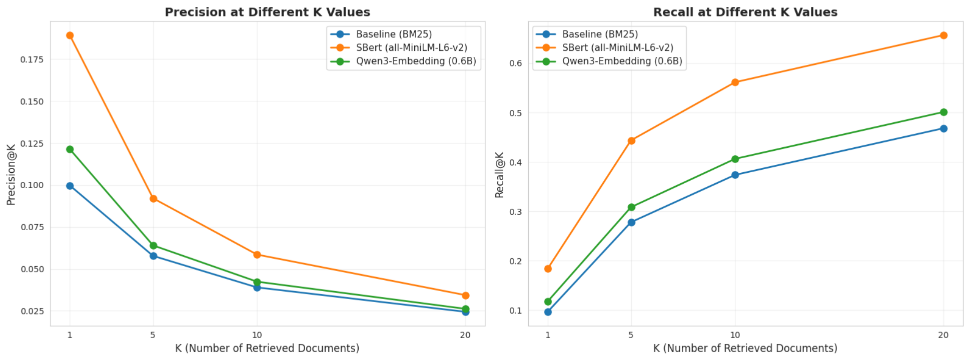

Figure 8: Precision and recall at different K values for MS MARCO passage evaluation.

Figure 9 shows the precision-recall trade-off at different K values.

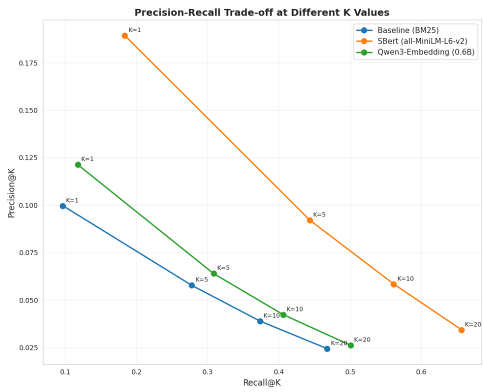

Figure 9: Precision-recall trade-off at different K values for MS MARCO passage evaluation.

Figure 10 shows a radar chart with all retrieval metrics.

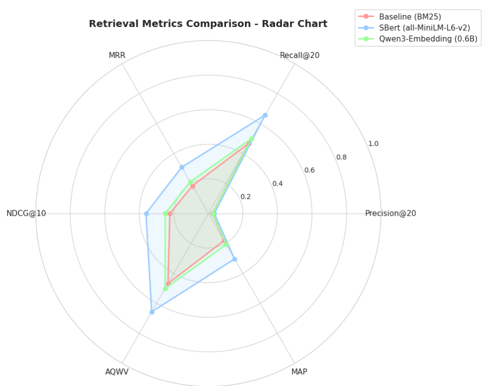

Figure 10: Radar chart showing all retrieval metrics for MS MARCO passage evaluation.

Figure 11 shows MRR and NDCG at different K values.

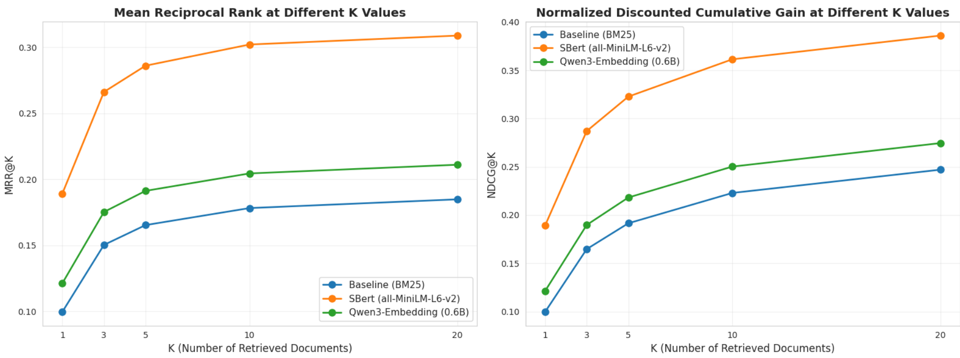

Figure 11: MRR and NDCG at different K values for MS MARCO passage evaluation.

Figure 12 shows the per-query metric differences.

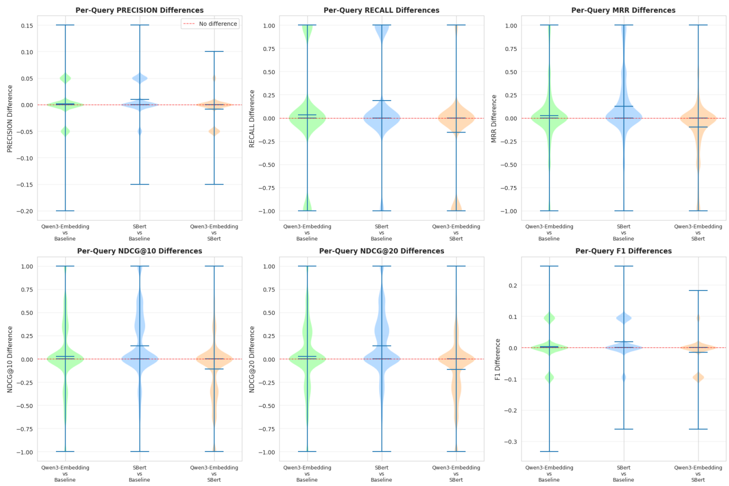

Figure 12: Per-query metric differences for MS MARCO passage evaluation.

Figure 13 compares generation metrics across all configurations. Here answer quality metrics and semantic similarity metrics are missing because this dataset does not contain gold answers.

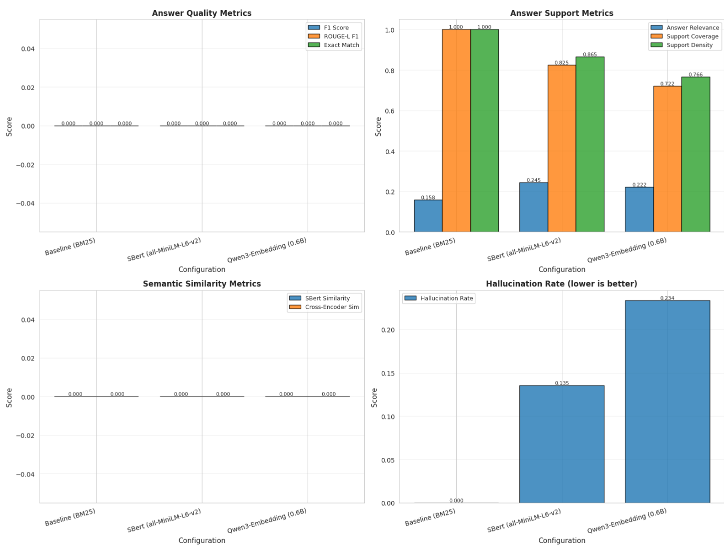

Figure 13: Generation metrics comparison for MS MARCO passage evaluation.

A.2 MS MARCO Question Answering

The MS MARCO QnA evaluation uses 10,000 sampled questions with both retrieval and generation assessment. This section provides comprehensive results including global metrics, statistical significance testing, generation metrics, and visualizations.

A.2.1 Global Retrieval Metrics

Table 10 presents the complete set of retrieval metrics across all three configurations.

Table 10: MS MARCO QnA: Global Retrieval Metrics Comparison

Metric	BM25	SBert	Qwen3
Precision@20	0.1338	0.1391	0.1093
Recall@1	0.0407	0.0413	0.0328
Recall@5	0.1441	0.1449	0.1113
Recall@10	0.2070	0.2107	0.1622
Recall@20	0.2676	0.2783	0.2186
F1 Score	0.1784	0.1855	0.1457
MRR	0.5043	0.5226	0.4253
NDCG@10	0.1645	0.2633	0.1760
NDCG@20	0.1802	0.2736	0.1860
MAP	0.1699	0.1679	0.1253
AQWV	0.2478	0.2579	0.2027

Table 11 presents the percentage improvements over baseline.

Table 11: MS MARCO QnA: Percentage Improvements Over BM25 Baseline (Retrieval Metrics)

Metric	SBert (%)	Qwen3 (%)
Precision@20	+3.96	-18.33
Recall@1	+1.47	-19.41
Recall@5	+0.55	-22.76
Recall@10	+1.79	-21.64
Recall@20	+4.00	-18.31
F1 Score	+3.98	-18.33
MRR	+3.63	-15.67
NDCG@10	+60.06	+6.99
NDCG@20	+51.83	+3.22
MAP	-1.18	-26.25
AQWV	+4.08	-18.20

A.2.2 Statistical Significance Testing

Table 12 presents detailed statistical significance testing results for SBert vs BM25.

Table 12: MS MARCO QnA: SBert vs BM25 Statistical Significance

Metric	SBert	BM25	Diff	Diff (%)	PAR p	Sig.
Precision	0.1855	0.1784	+0.0071	+3.98	<0.01	Yes
Recall	0.2783	0.2676	+0.0107	+4.00	<0.01	Yes
MRR	0.5226	0.5043	+0.0183	+3.63	<0.001	Yes
NDCG@10	0.2633	0.1645	+0.0988	+60.06	<0.001	Yes
NDCG@20	0.2736	0.1802	+0.0934	+51.83	<0.001	Yes
F1	0.1855	0.1784	+0.0071	+3.98	<0.01	Yes

Table 13 presents statistical significance testing results for Qwen3 vs BM25.

Table 13: MS MARCO QnA: Qwen3 vs BM25 Statistical Significance

Metric	Qwen3	BM25	Diff	Diff (%)	PAR p	Sig.
Precision	0.1457	0.1784	-0.0327	-18.33	<0.001	Yes
Recall	0.2186	0.2676	-0.0490	-18.32	<0.001	Yes
MRR	0.4253	0.5043	-0.0790	-15.67	<0.001	Yes
NDCG@10	0.1760	0.1645	+0.0115	+6.99	<0.05	Yes
NDCG@20	0.1860	0.1802	+0.0058	+3.22	0.16	No
F1	0.1457	0.1784	-0.0327	-18.33	<0.001	Yes

Table 14 presents statistical significance testing results for Qwen3 vs SBert.

Table 14: MS MARCO QnA: Qwen3 vs SBert Statistical Significance

Metric	Qwen3	SBert	Diff	Diff (%)	PAR p	Sig.
Precision	0.1457	0.1855	-0.0398	-21.46	<0.001	Yes
Recall	0.2186	0.2783	-0.0597	-21.45	<0.001	Yes
MRR	0.4253	0.5226	-0.0973	-18.62	<0.001	Yes
NDCG@10	0.1760	0.2633	-0.0873	-33.16	<0.001	Yes
NDCG@20	0.1860	0.2736	-0.0876	-32.02	<0.001	Yes
F1	0.1457	0.1855	-0.0398	-21.46	<0.001	Yes

A.2.3 Generation Metrics

Table 15 presents comprehensive generation quality metrics across all configurations.

Table 15: MS MARCO QnA: Generation Metrics Comparison

Metric	BM25	SBert	Qwen3
F1 Score	0.1994*	0.3660	0.3092
ROUGE-L F1	0.1726*	0.3253	0.2751
Exact Match	0.0156*	0.0342	0.0288
Answer Relevance	0.3782*	0.5789	0.5104
Support Coverage	1.0000*	0.7647	0.6659
Support Density	1.0000*	0.6891	0.6003
Hallucination Rate	0.0000*	0.1879	0.2844
SBert Similarity	0.5605*	0.7207	0.6196
Cross-Encoder Sim	0.5543*	0.7518	0.6431

* BM25 uses top-1 retrieval generation (not generative model)

A.2.4 Performance Visualization

Following are the evaluation graphs allowing for easier overview and evaluation of the many of the values already present in the preceding tables. For completeness we include all generated plots of the obtained results in the appendix for interested readers to interpret them themselves. The plots deemed most relevant are analyzed in the reports result section above, outside of the appendix.

Figure 14 shows precision and recall at different K values.

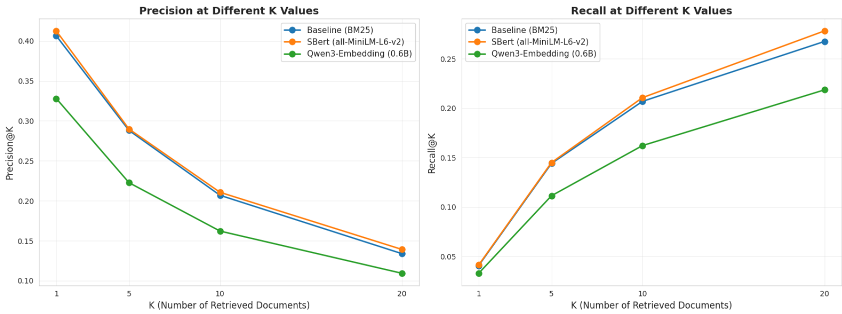

Figure 14: Precision and recall at different K values for MS MARCO QnA evaluation.

Figure 15 shows the precision-recall trade-off at different K values.

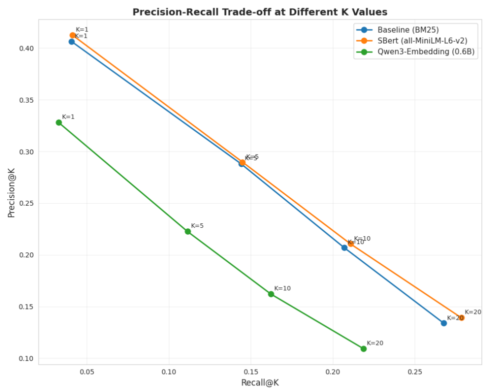

Figure 15: Precision-recall trade-off at different K values for MS MARCO QnA evaluation.

Figure 16 shows a radar chart with all retrieval metrics.

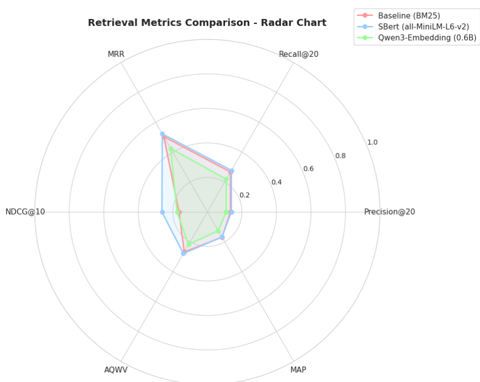

Figure 16: Radar chart showing all retrieval metrics for MS MARCO QnA evaluation.

Figure 17 shows MRR and NDCG at different K values.

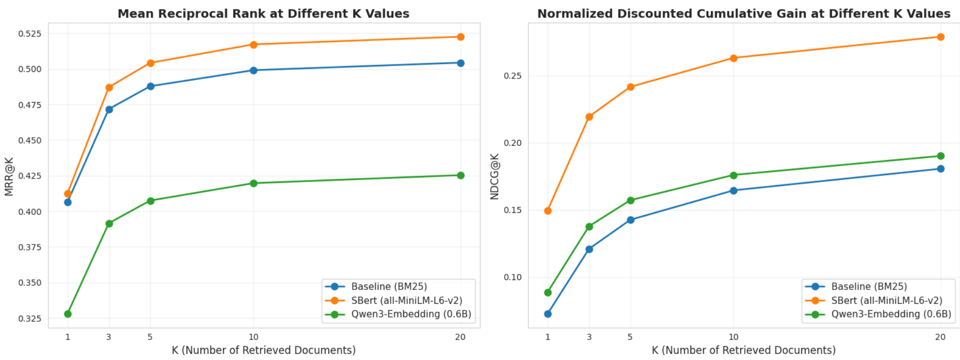

Figure 17: MRR and NDCG at different K values for MS MARCO QnA evaluation.

Figure 18 shows the per-query metric differences.

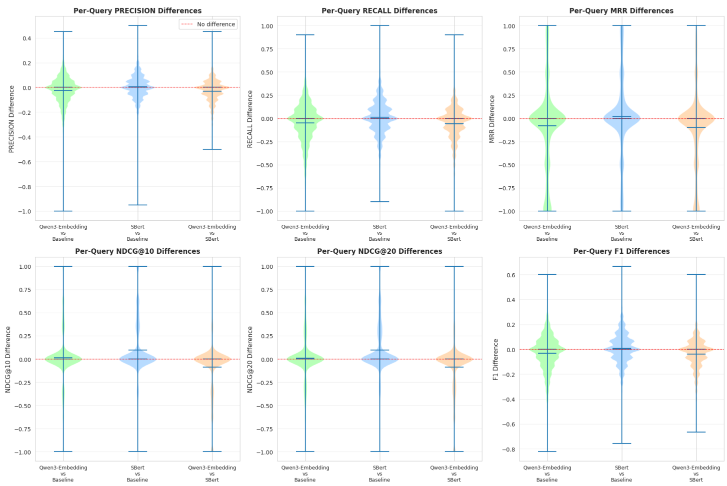

Figure 18: Per-query metric differences for MS MARCO QnA evaluation.

Figure 19 compares generation metrics across all configurations.

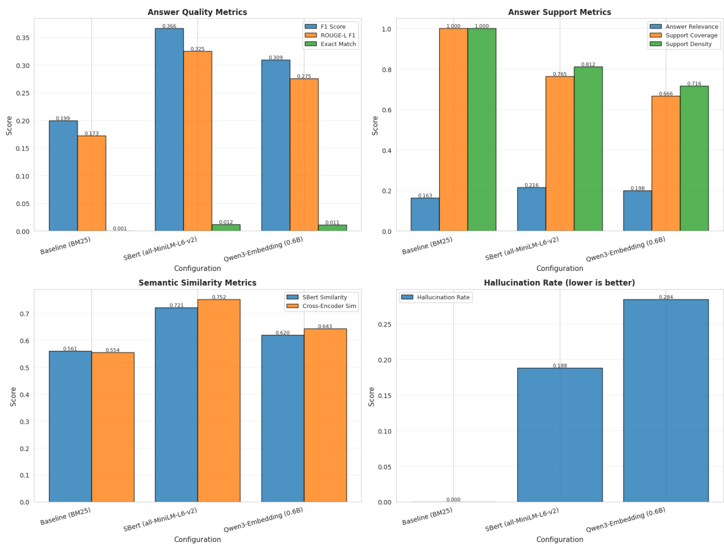

Figure 19: Generation metrics comparison for MS MARCO QnA evaluation.

A.3 OpenLifeLog QA

OpenLifeLog QA evaluation reveals distinct performance patterns for personal information retrieval, evaluated on 1,508 queries. This section provides comprehensive results including global metrics, statistical significance testing, generation metrics, and visualizations.

A.3.1 Global Retrieval Metrics

Table 16 presents the complete set of retrieval metrics across all three configurations.

Table 16: OpenLifeLog QA: Global Retrieval Metrics Comparison

Metric	BM25	SBert	Qwen3
Precision@20	0.0440	0.0119	0.0359
Recall@1	0.1280	0.0346	0.1379
Recall@5	0.2998	0.0777	0.2748
Recall@10	0.3983	0.1072	0.3397
Recall@20	0.5071	0.1463	0.4105
F1 Score	0.0770	0.0214	0.0634
MRR	0.2768	0.0770	0.2850
NDCG@10	0.2814	0.0730	0.2619
NDCG@20	0.3215	0.0883	0.2920
MAP	0.2369	0.0592	0.2201
AQWV	0.4793	0.1176	0.3825

Table 17 shows performance relative to BM25.

Table 17: OpenLifeLog QA: Percentage Improvements Over BM25 Baseline

Metric	SBert (%)	Qwen3 (%)
Precision@20	-72.95	-18.46
Recall@1	-72.97	+7.70
Recall@5	-74.09	-8.34
Recall@10	-73.09	-14.73
Recall@20	-71.15	-19.06
F1 Score	-72.19	-17.73
MRR	-72.17	+2.95
NDCG@10	-74.06	-6.91
NDCG@20	-72.54	-9.18
MAP	-75.00	-7.09
AQWV	-75.47	-20.21

A.3.2 Statistical Significance Testing

Table 18 presents detailed statistical significance testing results for SBert vs BM25.

Table 18: OpenLifeLog QA: SBert vs BM25 Statistical Significance

Metric	SBert	BM25	Diff	Diff (%)	PAR p	Sig.
Precision	0.0214	0.0770	-0.0556	-72.19	<0.001	Yes
Recall	0.1463	0.5071	-0.3608	-71.15	<0.001	Yes
MRR	0.0770	0.2768	-0.1998	-72.17	<0.001	Yes
NDCG@10	0.0730	0.2814	-0.2084	-74.06	<0.001	Yes
NDCG@20	0.0883	0.3215	-0.2332	-72.54	<0.001	Yes
F1	0.0214	0.0770	-0.0556	-72.19	<0.001	Yes

Table 19 presents statistical significance testing results for Qwen3 vs BM25.

Table 19: OpenLifeLog QA: Qwen3 vs BM25 Statistical Significance

Metric	Qwen3	BM25	Diff	Diff (%)	PAR p	Sig.
Precision	0.0634	0.0770	-0.0136	-17.73	<0.001	Yes
Recall	0.4105	0.5071	-0.0966	-19.05	<0.001	Yes
MRR	0.2850	0.2768	+0.0082	+2.96	0.447	No
NDCG@10	0.2619	0.2814	-0.0195	-6.93	0.044	Yes
NDCG@20	0.2920	0.3215	-0.0295	-9.18	0.006	Yes
F1	0.0634	0.0770	-0.0136	-17.73	<0.001	Yes

Table 20 presents statistical significance testing results for Qwen3 vs SBert.

Table 20: OpenLifeLog QA: Qwen3 vs SBert Statistical Significance

Metric	Qwen3	SBert	Diff	Diff (%)	PAR p	Sig.
Precision	0.0634	0.0214	+0.0420	+196.26	<0.001	Yes
Recall	0.4105	0.1463	+0.2642	+180.59	<0.001	Yes
MRR	0.2850	0.0770	+0.2080	+270.13	<0.001	Yes
NDCG@10	0.2619	0.0730	+0.1889	+258.77	<0.001	Yes
NDCG@20	0.2920	0.0883	+0.2037	+230.69	<0.001	Yes
F1	0.0634	0.0214	+0.0420	+196.26	<0.001	Yes

A.3.3 Generation Metrics

Table 21 presents comprehensive generation quality metrics across all configurations.

Table 21: OpenLifeLog QA: Generation Metrics Comparison

Metric	BM25	SBert	Qwen3
F1 Score	0.1276*	0.1487	0.1759
ROUGE-L F1	0.1146*	0.1472	0.1725
Exact Match	0.0159*	0.0159	0.0238
Answer Relevance	0.3420*	0.1557	0.2494
Support Coverage	1.0000*	0.3062	0.4331
Support Density	1.0000*	0.2512	0.4044
Hallucination Rate	0.0000*	0.7488	0.5956
SBert Similarity	0.2770*	0.2065	0.2751
Cross-Encoder Sim	0.0495*	0.0313	0.0876

* BM25 uses top-1 retrieval generation (not generative model)

A.3.4 Performance Visualization

Following are the evaluation graphs allowing for easier overview and evaluation of the many of the values already present in the preceding tables. For completeness we include all generated plots of the obtained results in the appendix for interested readers to interpret them themselves. The plots deemed most relevant are analyzed in the reports result section above, outside of the appendix.

Figure 20 shows precision and recall at different K values.

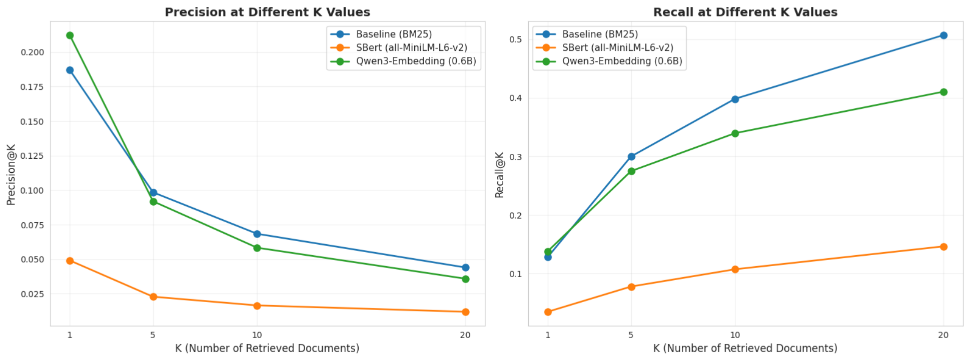

Figure 20: Precision and recall at different K values for OpenLifeLog QA evaluation.

Figure 21 shows the precision-recall trade-off at different K values.

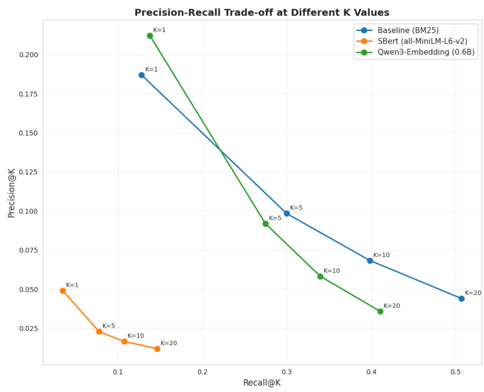

Figure 21: Precision-recall trade-off at different K values for OpenLifeLog QA evaluation.

Figure 22 shows a radar chart with all retrieval metrics.

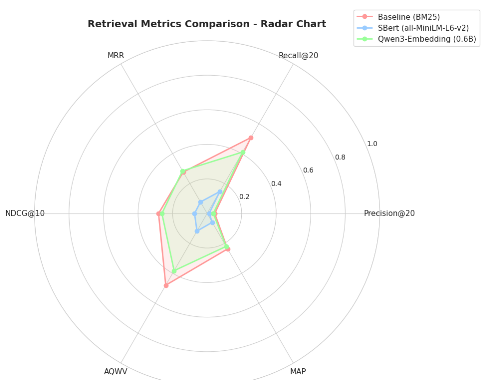

Figure 22: Radar chart showing all retrieval metrics for OpenLifeLog QA evaluation.

Figure 23 shows MRR and NDCG at different K values.

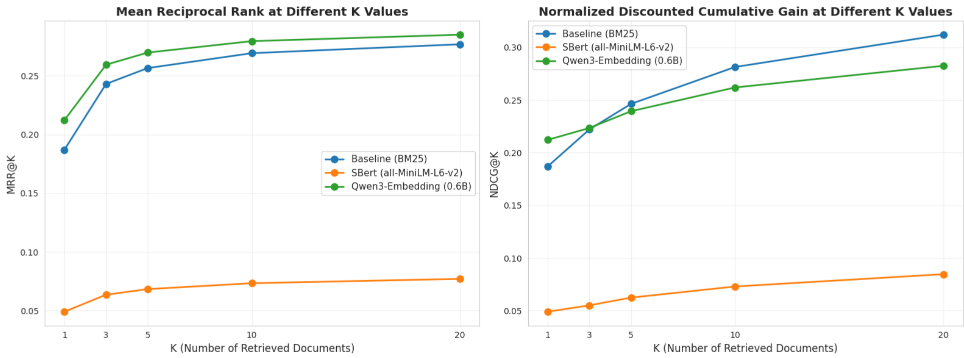

Figure 23: MRR and NDCG at different K values for OpenLifeLog QA evaluation.

Figure 24 shows the per-query metric differences.

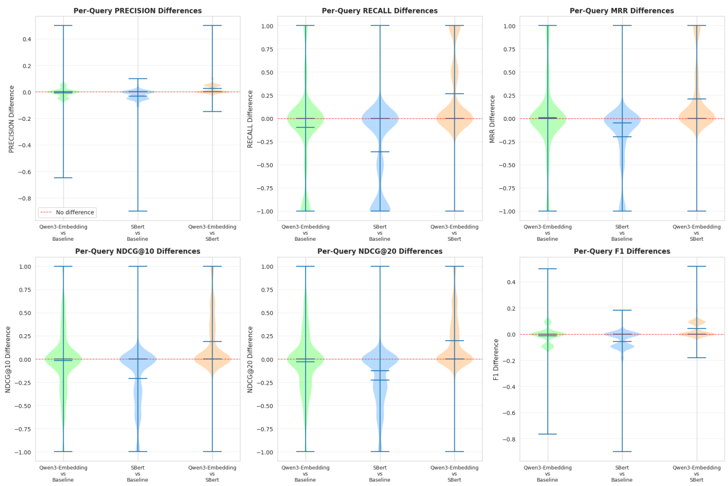

Figure 24: Per-query metric differences for OpenLifeLog QA evaluation.

Figure 25 compares generation metrics across all configurations.

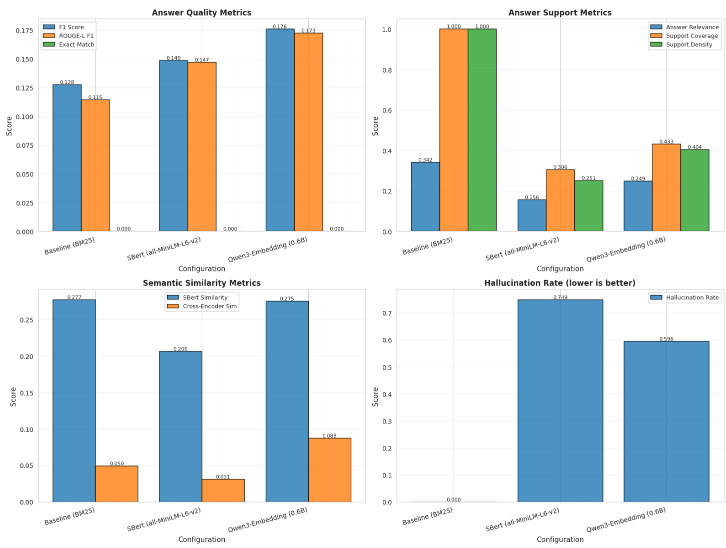

Figure 25: Generation metrics comparison for OpenLifeLog QA evaluation.

A.4 Timeline QA - Medium Configuration

For conciseness and because of critique for an overly long appendix we decided to remove the dense and sparse dataset versions of TimelineQA. If interested in these not so relevant results to our project feel free to look it up in the documentation of the project.

Timeline QA medium configuration represents moderate temporal event density, evaluated on 10,000 queries. This section provides comprehensive results including global metrics, statistical significance testing, generation metrics, and visualizations.

A.4.1 Global Retrieval Metrics

Table 22 presents the complete set of retrieval metrics across all three configurations.

Table 22: Timeline QA Medium: Global Retrieval Metrics Comparison

Metric	BM25	SBert	Qwen3
Precision@20	0.0450	0.0190	0.0372
Recall@1	0.4882	0.1390	0.3864
Recall@5	0.6818	0.2542	0.5810
Recall@10	0.7604	0.3139	0.6634
Recall@20	0.9006	0.3796	0.7446
F1 Score	0.0858	0.0362	0.0709
MRR	0.5901	0.1929	0.4767
NDCG@10	0.6232	0.2181	0.5169
NDCG@20	0.6584	0.2461	0.5587
MAP	0.5901	0.1929	0.4767
AQWV	0.8770	0.3553	0.7208

Table 23 shows performance metrics.

Table 23: Timeline QA Medium: Percentage Improvements Over BM25 Baseline

Metric	SBert (%)	Qwen3 (%)
Precision@20	-57.78	-17.33
Recall@1	-71.53	-20.85
Recall@5	-62.72	-14.78
Recall@10	-58.72	-12.76
Recall@20	-57.85	-17.32
F1 Score	-57.81	-17.37
MRR	-67.31	-19.22
NDCG@10	-65.01	-17.06
NDCG@20	-62.62	-15.14
MAP	-67.31	-19.22
AQWV	-59.50	-17.82

A.4.2 Statistical Significance Testing

Table 24 presents detailed statistical significance testing results for SBert vs BM25.

Table 24: Timeline QA Medium: SBert vs BM25 Statistical Significance

Metric	SBert	BM25	Diff	Diff (%)	PAR p	Sig.
Precision	0.0362	0.0858	-0.0496	-57.81	<0.001	Yes
Recall	0.3796	0.9006	-0.5210	-57.85	<0.001	Yes
MRR	0.1929	0.5901	-0.3972	-67.31	<0.001	Yes
NDCG@10	0.2181	0.6232	-0.4051	-65.01	<0.001	Yes
NDCG@20	0.2461	0.6584	-0.4123	-62.62	<0.001	Yes
F1	0.0362	0.0858	-0.0496	-57.81	<0.001	Yes

Table 25 presents statistical significance testing results for Qwen3 vs BM25.

Table 25: Timeline QA Medium: Qwen3 vs BM25 Statistical Significance

Metric	Qwen3	BM25	Diff	Diff (%)	PAR p	Sig.
Precision	0.0709	0.0858	-0.0149	-17.37	<0.001	Yes
Recall	0.7446	0.9006	-0.1560	-17.32	<0.001	Yes
MRR	0.4767	0.5901	-0.1134	-19.22	<0.001	Yes
NDCG@10	0.5169	0.6232	-0.1063	-17.06	<0.001	Yes
NDCG@20	0.5587	0.6584	-0.0997	-15.14	<0.001	Yes
F1	0.0709	0.0858	-0.0149	-17.37	<0.001	Yes

Table 26 presents statistical significance testing results for Qwen3 vs SBert.

Table 26: Timeline QA Medium: Qwen3 vs SBert Statistical Significance

Metric	Qwen3	SBert	Diff	Diff (%)	PAR p	Sig.
Precision	0.0709	0.0362	+0.0347	+95.86	<0.001	Yes
Recall	0.7446	0.3796	+0.3650	+96.15	<0.001	Yes
MRR	0.4767	0.1929	+0.2838	+147.12	<0.001	Yes
NDCG@10	0.5169	0.2181	+0.2988	+137.01	<0.001	Yes
NDCG@20	0.5587	0.2461	+0.3126	+127.00	<0.001	Yes
F1	0.0709	0.0362	+0.0347	+95.86	<0.001	Yes

A.4.3 Generation Metrics

Table 27 presents comprehensive generation quality metrics across all configurations.

Table 27: Timeline QA Medium: Generation Metrics Comparison

Metric	BM25	SBert	Qwen3
F1 Score	0.4882*	0.1390	0.3864
ROUGE-L F1	0.4882*	0.1390	0.3864
Exact Match	0.4882*	0.1390	0.3864
Answer Relevance	0.9764*	0.3475	0.7728
Support Coverage	1.0000*	0.9281	0.9741
Support Density	1.0000*	0.9281	0.9741
Hallucination Rate	0.0000*	0.0719	0.0259
SBert Similarity	0.9162*	0.5890	0.8294
Cross-Encoder Sim	0.9623*	0.6523	0.9009

* BM25 uses top-1 retrieval generation (not generative model)

A.4.4 Performance Visualization

Following are the evaluation graphs allowing for easier overview and evaluation of the many of the values already present in the preceding tables. For completeness we include all generated plots of the obtained results in the appendix for interested readers to interpret them themselves. The plots deemed most relevant are analyzed in the reports result section above, outside of the appendix.

Figure 26 shows precision and recall at different K values.

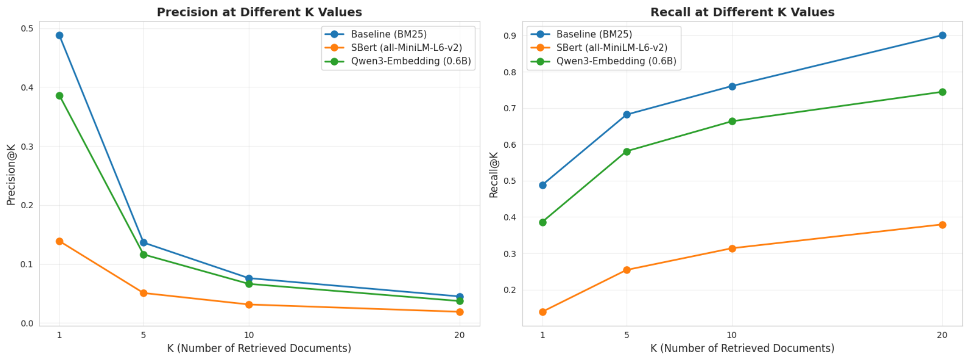

Figure 26: Precision and recall at different K values for Timeline QA medium configuration.

Figure 27 shows the precision-recall trade-off at different K values.

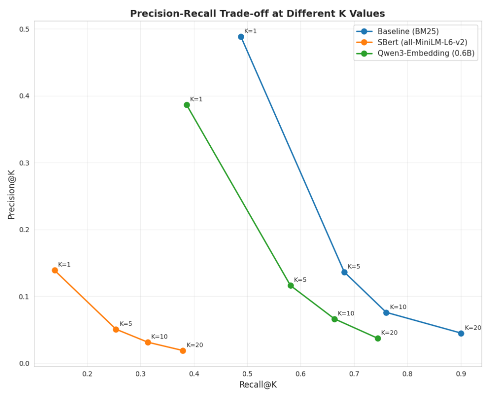

Figure 27: Precision-recall trade-off at different K values for Timeline QA medium configuration.

Figure 28 shows a radar chart with all retrieval metrics.

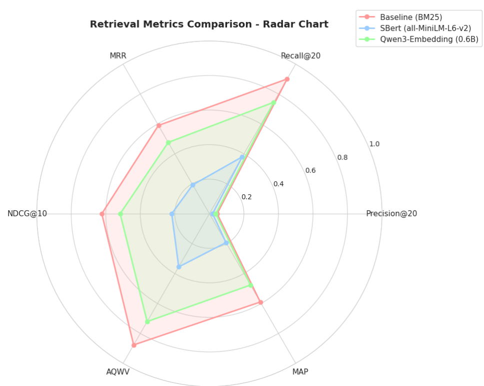

Figure 28: Radar chart showing all retrieval metrics for Timeline QA medium configuration.

Figure 29 shows MRR and NDCG at different K values.

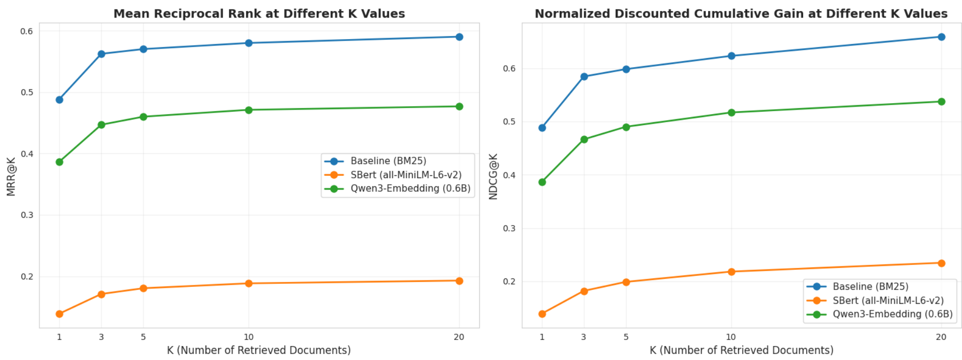

Figure 29: MRR and NDCG at different K values for Timeline QA medium configuration.

Figure 30 shows the per-query metric differences.

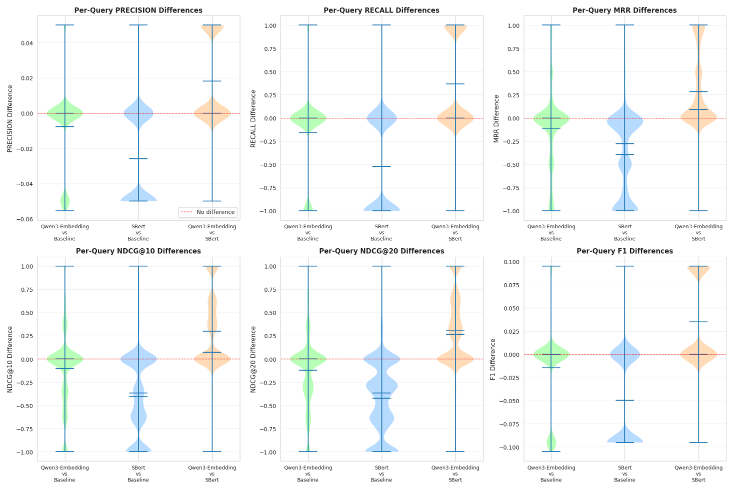

Figure 30: Per-query metric differences for Timeline QA medium configuration.

Figure 31 compares generation metrics across all configurations.

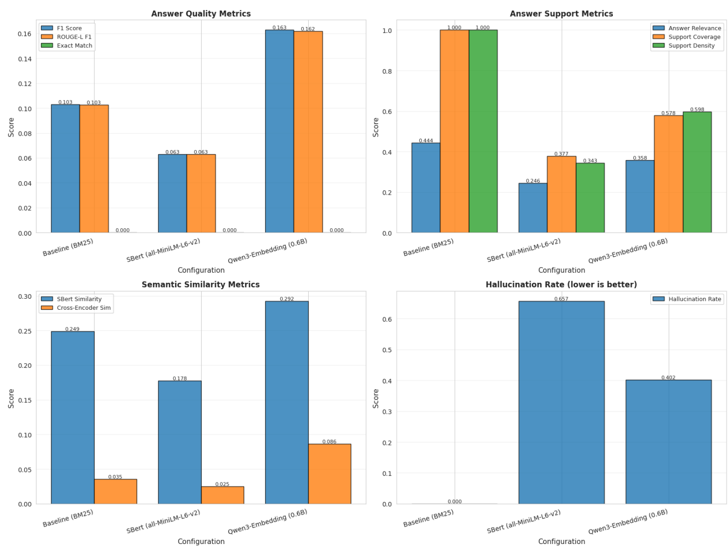

Figure 31: Generation metrics comparison for Timeline QA medium configuration.